

1 Structured summary: | Optimal Allocation of Observations
2 across Run-In, Treatment, and Common Close Phases in
3 Longitudinal Clinical Trials

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27 *This document is a structured, section-by-section summary of the key points argued in the*
28 *companion manuscript, “| Optimal Allocation of Observations across Run-In, Treatment, and*
29 *Common Close Phases in Longitudinal Clinical Trials.” It is provided as a supplementary*

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30 reading aid and does not stand on its own; all notation, derivations, simulation detail, and
31 citations are in the main manuscript.

32

33 1 Introduction

34 **Additional run-in and common close observations both improve efficiency but**
35 **with diminishing or J_1 -dependent returns.**

- 36 • Run-in observations appended to a fixed treatment phase improve efficiency with dimin-
37 ishing returns.
- 38 • Most of the attainable gain comes from the first two run-in observations; the marginal
39 value falls quickly thereafter.
- 40 • Common close gains depend on the number of treatment observations J_1 .

41 **The efficiency results motivate an allocation question left open by Paper 1.**

- 42 • Trials operate under constraints on total visits or total duration.
- 43 • Paper 1 characterizes efficiency but not the optimal phase split.
- 44 • The question is how to distribute observations across the three phases.

45 **Prior work addresses related design questions but not optimal allocation across**
46 **run-in, treatment, and common close phases.**

- 47 • Existing optima target time-point positioning or general design guidelines, not the run-in
48 setting.
- 49 • Multiple pre-treatment measurements are known to benefit ANCOVA-analyzed designs.
- 50 • A substantial optimal-design literature treats measurement schedules for mixed models
51 through the Fisher information matrix, but does not address a phase structure in which
52 one phase carries its own nuisance parameter.
- 53 • Available software computes slope-comparison power but does not optimize the phase
54 allocation.

55 1.1 Fixed total visits

56 **The fixed-visit problem minimizes the treatment-effect variance over the phase**
57 **split under a total-count constraint.**

- 58 • The budget J excludes baseline and must be split as $J_0 + J_1 + J_2 = J$.
- 59 • Feasibility requires $J_0 \geq 0$, $J_1 \geq 1$, and $J_2 \geq 0$.
- 60 • The objective is the $\gamma\gamma$ entry of $(X'\Sigma^{-1}X)^{-1}$.

61 1.2 Fixed total duration

62 **The fixed-duration problem instead constrains total trial length and may also**
63 **optimize observation timing.**

- 64 • Duration is fixed via $T = (J_0 + J_1 + J_2) \cdot t$, so the count constraint becomes $J_0 + J_1 + J_2 =$
65 T/t .
- 66 • The variance $\text{Var}_1(\hat{\gamma})$ is minimized subject to this constraint.
- 67 • A general version permits non-equally-spaced observations and optimizes over the ob-
68 servation times.

69 1.3 The $J_0 = 1$ anomaly

70 **In an intermediate signal-to-noise band the first run-in observation is less valuable**
71 **than the second because it introduces a new model parameter.**

- 72 • At $J_1 = 2$, $J_2 = 0$, $\sigma = t = 1$, $\Delta V_1 < \Delta V_2$ holds for σ_b/σ between approximately 0.53
73 and 1.51, and fails outside this band.
- 74 • Adding one run-in observation expands the model from two to three parameters, con-
75 suming a degree of freedom.
- 76 • The second run-in observation adds no parameter and delivers pure information gain.

77 **The single-run-in penalty parallels a known ANCOVA result and yields a regime-**
78 **specific design rule.**

- 79 • @frison1992 found a single pre-treatment ANCOVA covariate can reduce power at low
80 pre-post correlation.
- 81 • The same degree-of-freedom cost applies in the run-in setting.
- 82 • Within the intermediate band of Table
83 reftab:anomaly-boundary, designers who add run-in observations should plan at least
84 two rather than exactly one.

85 1.4 Correlation structure governs diminishing returns

86 **Under compound symmetry diminishing returns stem from the information ma-**
87 **trix, whereas under AR(1) declining correlation accelerates them.**

- 88 • The run-in change-score correlation is constant in J_0 under compound symmetry.
- 89 • Diminishing returns for $J_0 \geq 2$ therefore arise from the information matrix structure.
- 90 • Under AR(1) the correlation falls with temporal distance, yielding faster diminishing
91 returns from distant run-in observations.

92 2 Optimal allocation for fixed J

93 **Under a fixed total budget the optimum never allocates observations to the run-in**
94 **phase.**

- In every cell of Table reftab:optimal-grid, $J_0^* = 0$; the budget splits between treatment and common close.
- The run-in phase adds the pre-randomization slope parameter δ without contributing to the treatment contrast, so a run-in visit is dominated by a treatment or close visit whenever visits are fungible.
- Paper 1's run-in gains are not contradicted: they arise when run-in visits are additional, appended to a fixed treatment phase, not reallocated from it.

3 When is the common close phase beneficial?

The common close phase aids estimation only indirectly, through the random slope, so the meaningful question is the fixed-budget trade-off.

- Common close observations inform the random slope b_i ; they do not contribute to the treatment comparison directly.
- Added to a fixed design, a common close observation reduced $\text{Var}(\hat{\gamma})$ in every configuration examined ($J_1 = 1, \dots, 4$, σ_b/σ down to 10^{-3}), so no benefit threshold exists for free observations.
- The design-relevant question is whether a budgeted visit is better spent on the common close than on the treatment phase.

3.1 Crossover as a function of the budget J

The crossover ratio falls as the budget grows, and above it the close phase absorbs an increasing share of the budget.

- The crossover decreases from 0.81 at $J = 3$ to 0.30 at $J = 8$.
- Above the crossover the full optimizer allocates a growing share to the close, reaching an even split $(0, 4, 4)$ at $J = 8$ at every examined ratio from 0.6 upward.
- Even at $\sigma_b/\sigma = 0.32$ the optimum at $J = 8$ is $(0, 5, 3)$, so the close enters at low ratios once the budget is large.

3.2 General time points

Maximally separated time points minimize the slope-estimation variance, consistent with the classical extremes result.

- Within the raw-outcome model, the variance is minimized when $t_1 t_2$ is most negative.
- This occurs when the two time points are maximally separated.
- It is consistent with the known optimality of observations at the extremes of follow-up for slope estimation.

127 3.3 Motivation

128 **The averaging logic that motivates run-in can be applied symmetrically at the**
129 **end of treatment to reduce endpoint measurement error.**

- 130 • Run-in averages pre-randomization observations to cut the measurement-error contribu-
131 tion to the baseline slope.
- 132 • Replicated measures raise power with diminishing returns past 20-50 replications.
- 133 • Closely spaced replicates of the final visit average out endpoint measurement error under
134 the same treatment condition.

135 **Replicating an endpoint or baseline visit scales down its residual variance while**
136 **leaving the random-slope contribution intact.**

- 137 • Replicating the last visit n_{rep} times over a short interval gives effective residual variance
138 σ^2/n_{rep} .
- 139 • The random slope contribution $\sigma_b^2 t^2$ is essentially unchanged.
- 140 • Replicating the baseline visit similarly reduces σ^2 in the change-score covariance.

141 **Replicated endpoints differ fundamentally from common close observations in**
142 **timing and the information they provide.**

- 143 • Common close observations occur at new time points after treatment stops.
- 144 • They inform the off-treatment slope.
- 145 • Replicated endpoints sit at the same time point as the last treatment visit and only
146 reduce measurement error.

147 3.4 Variance formula with replication

148 **Replication enters only through modified entries of R , leaving the Woodbury**
149 **computation otherwise unchanged.**

- 150 • The diagonal entry for a replicated visit uses σ^2/n_{rep} in place of σ^2 .
- 151 • Off-diagonal entries reflecting shared baseline error use $\sigma^2/n_{\text{rep,pre}}$.
- 152 • The remaining Woodbury computation proceeds identically.

153 **Endpoint replication helps most when measurement error dominates and little**
154 **when slope variability does.**

- 155 • The benefit is largest at small σ_b/σ .
- 156 • At large $\sigma_b/\sigma > 2$ the gain is modest.
- 157 • Variance is then dominated by $\sigma_b^2 t^2$, which replication cannot reduce.

158 3.5 Replication vs additional run-in observations

159 **A fixed budget of extra measurements can be allocated either to run-in observa-**
160 **tions or to endpoint replicates.**

- 161 • Run-in observations sit at distinct time points separated by t .

- Endpoint replicates are closely spaced at one time point.
- Replicates provide measurement-error reduction only, not slope information.

Whether run-in or endpoint replication wins depends on the signal-to-noise ratio and on the number of extra measurements.

- At $\sigma_b/\sigma = 1$, a single extra measurement is better spent on a replicate (variance 2.73 versus 2.82); run-in wins from the second extra measurement onward.
- Run-in observations supply information about the individual slope b_i , not only measurement-error reduction, but a lone run-in observation pays the single-run-in penalty of Section 3.
- At $\sigma_b/\sigma = 0.5$, replication won at every budget examined (one to four extra measurements) because measurement error dominates the variance.

3.6 Combined replication at both ends

Replicating both terminal visits yields symmetric measurement-error reduction.

- Baseline replication reduces σ^2 at the start of follow-up.
- Endpoint replication reduces σ^2 at the end of treatment.
- Together they lower measurement error symmetrically at both ends.

3.7 Pattern-mixture approach

Dropout is accommodated by a pattern-mixture model that combines stratum-specific sample sizes.

- The approach follows @frost2008 Section 2.3 and related pattern-mixture work.
- Participants are stratified by their observed visit pattern.
- The overall requirement is a weighted average across strata.

Strata are truncated designs under monotone dropout, and the formula is a conservative approximation that exact treatments improve upon.

- Here p_s is the proportion with pattern s and N_s the size needed if all had pattern s .
- Under monotone dropout, the pattern after post-baseline visit j is the truncated design retaining the first j post-baseline observations; participants with no post-baseline visit contribute no slope-contrast information ($N_s = \infty$).
- @hu2021 derive exact formulas under monotone dropout, more rigorous and less conservative when dropout is informative about the random slope.

3.8 AD neuroimaging trial

The AD example evaluates optimal allocation under MIRIAD parameters across several budgets and dropout scenarios.

- Parameters are $\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, and $t = 1$ year.

- Allocation is compared for $J = 4, 5, 6$ total observations.
- Both with-dropout and without-dropout cases are considered.

4 Discussion

Under a fixed visit budget the optimum places nothing in the run-in phase, which reconciles rather than contradicts the run-in literature.

- In every configuration examined the optimizer sets $J_0^* = 0$ and splits the budget between treatment and common close.
- Run-in visits pay for the pre-randomization slope parameter δ while informing the treatment contrast only indirectly, so a fungible visit is always better spent elsewhere.
- Paper 1's efficiency gains apply when run-in visits are additional; the two framings answer different design questions.

The signal-to-noise ratio and the budget jointly determine how much of the budget the common close should absorb.

- Reallocating one visit from treatment to common close pays once σ_b/σ exceeds a crossover that falls from 0.81 at $J = 3$ to 0.30 at $J = 8$.
- Above the crossover the close phase absorbs a growing share, reaching an even split at $J = 8$ in the settings examined.
- Designers should estimate the slope-to-noise ratio from pilot data before fixing the phase split.

Two refinements, unequal spacing and endpoint replication, offer further efficiency without changing the phase counts.

- In the raw-outcome special case of Section 6, spacing two observations toward the extremes of the window minimizes the slope-estimation variance; the general change-score case remains open.
- Replicating the endpoint or baseline visit reduces measurement error and is most valuable when measurement error dominates.
- At $\sigma_b/\sigma = 1$, a single extra measurement favors an endpoint replicate; run-in observations at new time points win from the second extra measurement onward.

The allocation rules are derived under idealizing assumptions whose relaxation is the natural next step.

- The optima assume equally spaced observations, a linear trajectory, and compound-symmetric residuals.
- Dropout is handled by a pattern-mixture approximation over monotone truncation strata, which the exact treatment of @hu2021 would refine.
- Under AR(1) residuals the diminishing returns to distant run-in observations are steeper, shifting the optimum toward fewer, closer run-in observations.

232 5 Conflict of interest

233 **The author reports no conflicts of interest.**

- 234 • No financial conflicts are present.
- 235 • No non-financial conflicts are present.
- 236 • No competing interests bear on this work.

237 6 Funding

238 **The work received no specific funding.**

- 239 • No external grant supported the study.
- 240 • No internal grant supported the study.
- 241 • No sponsor influenced the analysis.

242 7 Data availability statement

243 **The implementing package, tests, and shared bibliography are openly available**
244 **in the project repository.**

- 245 • Allocation routines reside in allocation.R and replicated.R; design.R builds the design
246 matrices they rely on.
- 247 • Validation tests and the four-paper bibliography are included.
- 248 • Specific paths appear in the Reproducibility section below.

249 8 Reproducibility

250 **The manuscript build relies on a small, fixed set of R packages.**

- 251 • Rendering used R 4.4.x.
- 252 • The in-package runinpower source and tinytest provide computation and testing.
- 253 • rmarkdown and bookdown drive the manuscript build.

254 **All results are deterministic.**

- 255 • Every table and figure is a deterministic evaluation of the variance formulas.
- 256 • No Monte Carlo simulation appears in this manuscript.
- 257 • No random seed is therefore required.

258 **Source code, companion manuscripts, and the shared bibliography are located at**
259 **documented repository paths.**

- 260 • The R package source is under R/.
- 261 • Companion manuscripts (Papers 1, 2, 4) are under analysis/report/{01,02,04}-*/.

- 262 • This manuscript source and the shared bibliography are under analysis/report/03-
263 allocation/.